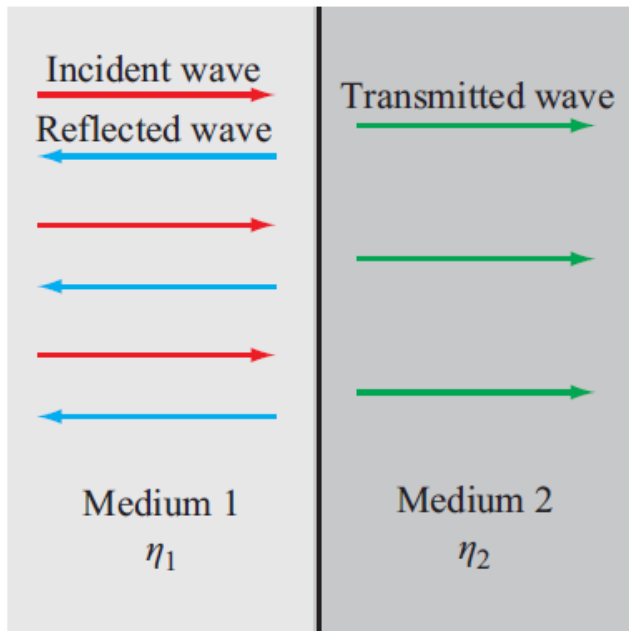
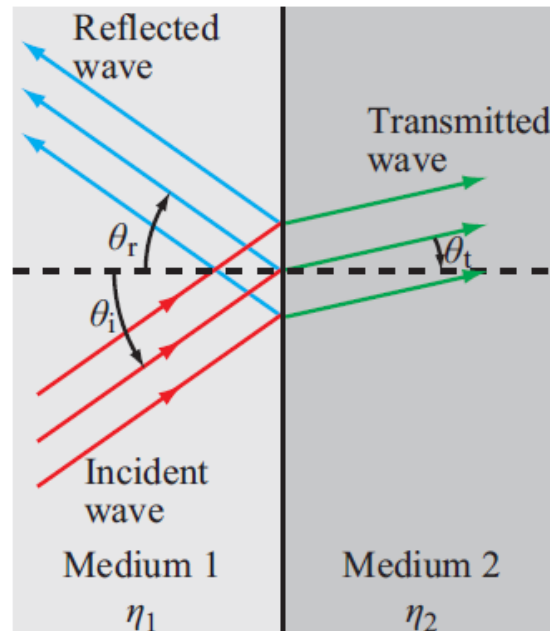


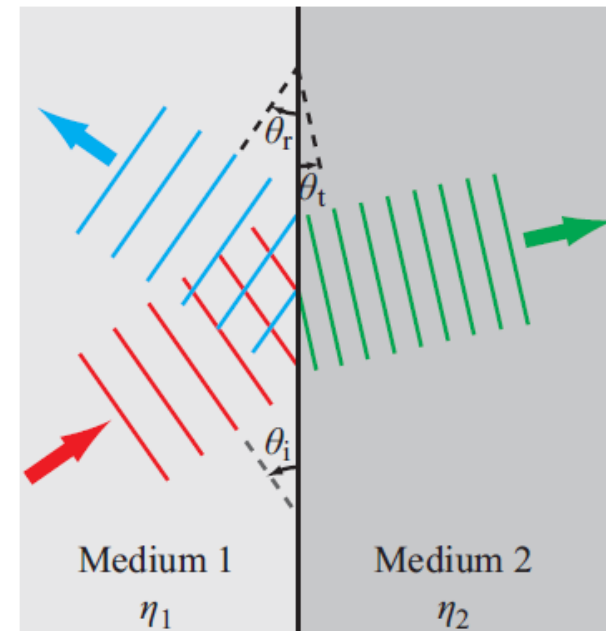
Normal and Oblique Incidence



(a) Normal incidence



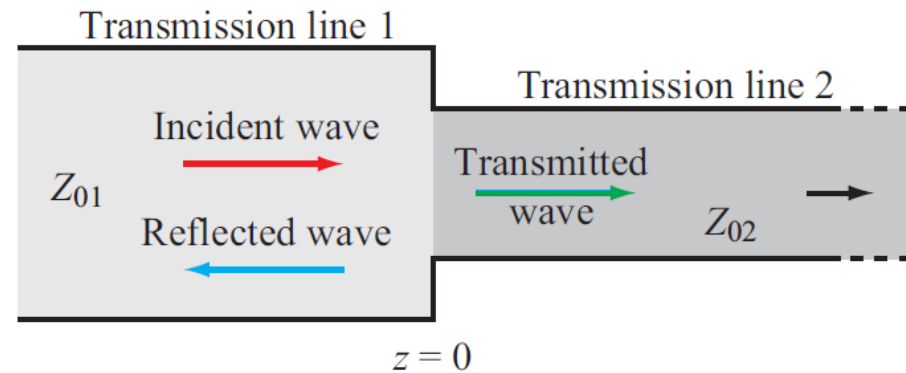
(b) Ray representation of oblique incidence



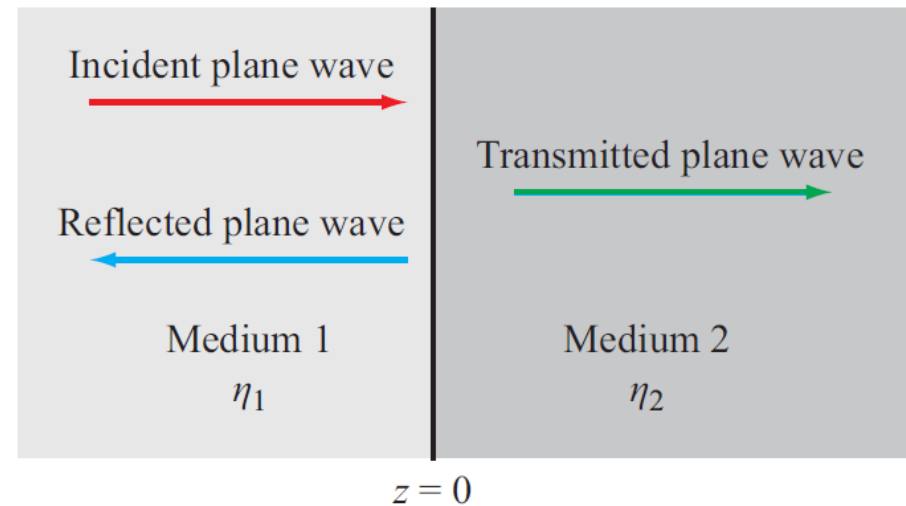
(c) Wavefront representation of oblique incidence

Normal Incidence

We can use all the transmission-line concepts and techniques to analyze plane wave reflection and transmission at interfaces between dissimilar media



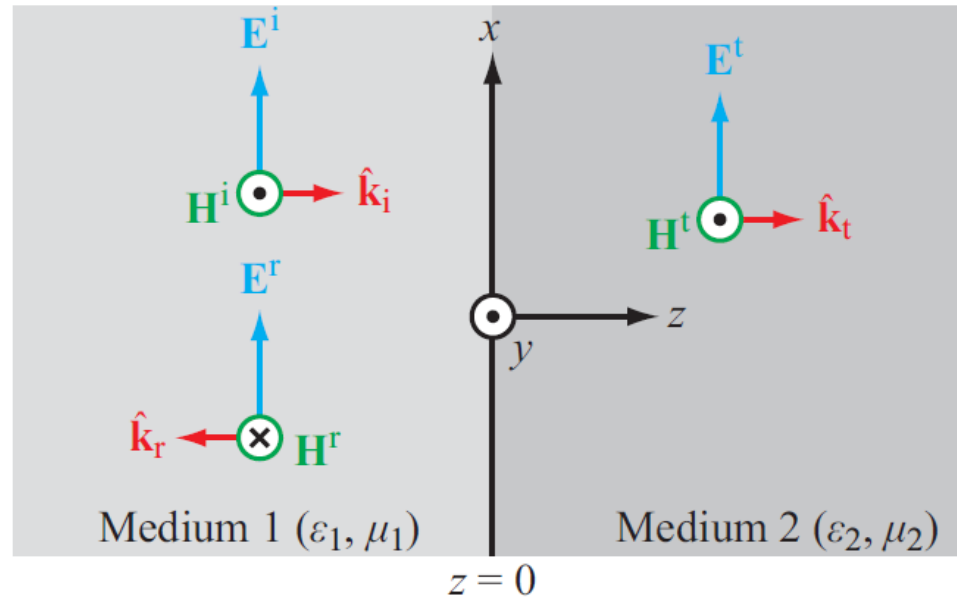
(a) Boundary between transmission lines



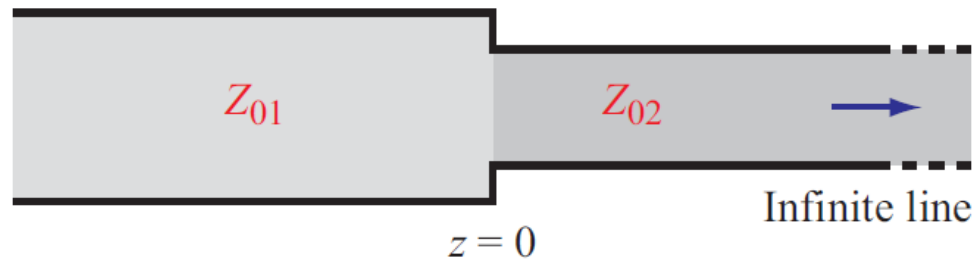
(b) Boundary between different media

Figure 8-2: Discontinuity between two different transmission lines is analogous to that between two dissimilar media.

Modeling Normal Incidence



(a) Boundary between dielectric media



(b) Transmission-line analogue

Individual Waves

Incident Wave

$$\tilde{\mathbf{E}}^i(z) = \hat{\mathbf{x}} E_0^i e^{-jk_1 z},$$

$$\tilde{\mathbf{H}}^i(z) = \hat{\mathbf{z}} \times \frac{\tilde{\mathbf{E}}^i(z)}{\eta_1} = \hat{\mathbf{y}} \frac{E_0^i}{\eta_1} e^{-jk_1 z}.$$

Reflected Wave

$$\tilde{\mathbf{E}}^r(z) = \hat{\mathbf{x}} E_0^r e^{jk_1 z},$$

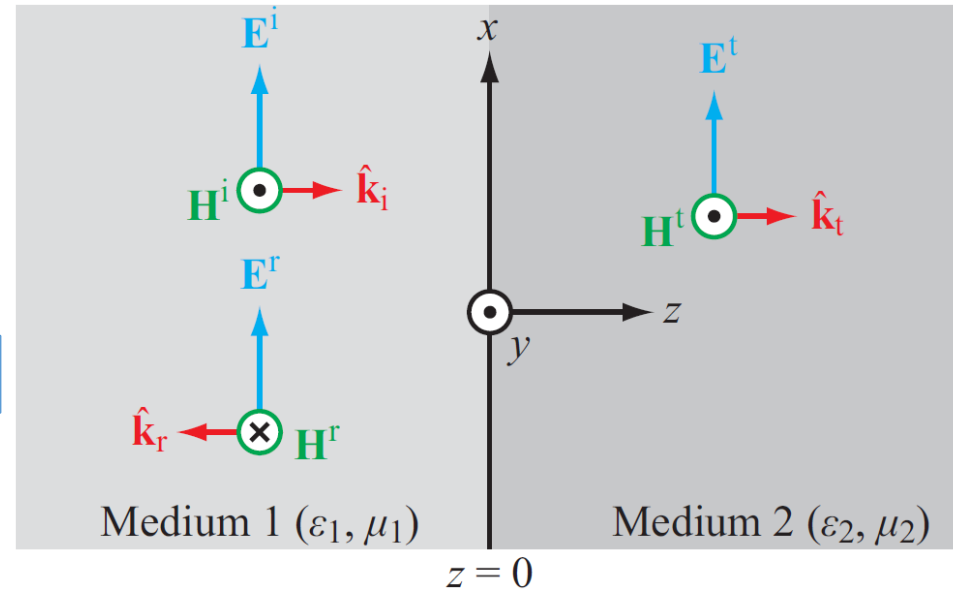
$$\tilde{\mathbf{H}}^r(z) = (-\hat{\mathbf{z}}) \times \frac{\tilde{\mathbf{E}}^r(z)}{\eta_1} = -\hat{\mathbf{y}} \frac{E_0^r}{\eta_1} e^{jk_1 z}.$$

Transmitted Wave

$$\tilde{\mathbf{E}}^t(z) = \hat{\mathbf{x}} E_0^t e^{-jk_2 z},$$

$$\tilde{\mathbf{H}}^t(z) = \hat{\mathbf{z}} \times \frac{\tilde{\mathbf{E}}^t(z)}{\eta_2} = \hat{\mathbf{y}} \frac{E_0^t}{\eta_2} e^{-jk_2 z}.$$

Note – sign



Note + sign

Note – sign

Boundary Conditions

Total fields

Medium 1

$$\begin{aligned}\tilde{\mathbf{E}}_1(z) &= \tilde{\mathbf{E}}^i(z) + \tilde{\mathbf{E}}^r(z) \\ &= \hat{\mathbf{x}}(E_0^i e^{-jk_1 z} + E_0^r e^{jk_1 z}),\end{aligned}\quad (8.4a)$$

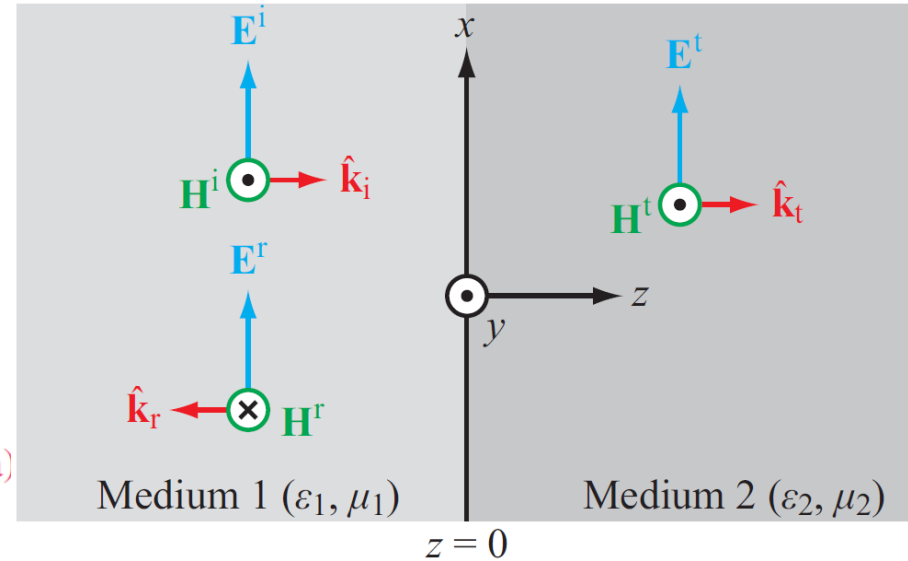
$$\begin{aligned}\tilde{\mathbf{H}}_1(z) &= \tilde{\mathbf{H}}^i(z) + \tilde{\mathbf{H}}^r(z) \\ &= \hat{\mathbf{y}} \frac{1}{\eta_1} (E_0^i e^{-jk_1 z} - E_0^r e^{jk_1 z}).\end{aligned}\quad (8.4b)$$

With only the transmitted wave present in medium 2, the total fields are

Medium 2

$$\tilde{\mathbf{E}}_2(z) = \tilde{\mathbf{E}}^t(z) = \hat{\mathbf{x}} E_0^t e^{-jk_2 z}, \quad (8.9a)$$

$$\tilde{\mathbf{H}}_2(z) = \tilde{\mathbf{H}}^t(z) = \hat{\mathbf{y}} \frac{E_0^t}{\eta_2} e^{-jk_2 z}. \quad (8.9b)$$



Solution gives:

$$E_0^r = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right) E_0^i = \Gamma E_0^i,$$

$$E_0^t = \left(\frac{2\eta_2}{\eta_2 + \eta_1} \right) E_0^i = \tau E_0^i,$$

where

$$\begin{aligned}\Gamma &= \frac{E_0^r}{E_0^i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (\text{normal incidence}), \\ \tau &= \frac{E_0^t}{E_0^i} = \frac{2\eta_2}{\eta_2 + \eta_1} \quad (\text{normal incidence}).\end{aligned}$$

At the boundary $z = 0$:

$$\tilde{\mathbf{E}}_1(0) = \tilde{\mathbf{E}}_2(0) \quad \text{or} \quad E_0^i + E_0^r = E_0^t,$$

$$\tilde{\mathbf{H}}_1(0) = \tilde{\mathbf{H}}_2(0) \quad \text{or} \quad \frac{E_0^i}{\eta_1} - \frac{E_0^r}{\eta_1} = \frac{E_0^t}{\eta_2}.$$

Reflection and Transmission Coefficients

$$\Gamma = \frac{E_0^r}{E_0^i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (\text{normal incidence}),$$
$$\tau = \frac{E_0^t}{E_0^i} = \frac{2\eta_2}{\eta_2 + \eta_1} \quad (\text{normal incidence}).$$

Similar form as for
transmission lines

$$\tau = 1 + \Gamma \quad (\text{normal incidence}).$$

$$\Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} \quad (\text{nonmagnetic media}).$$

Analogy of Normal Incidence to Transmission Lines

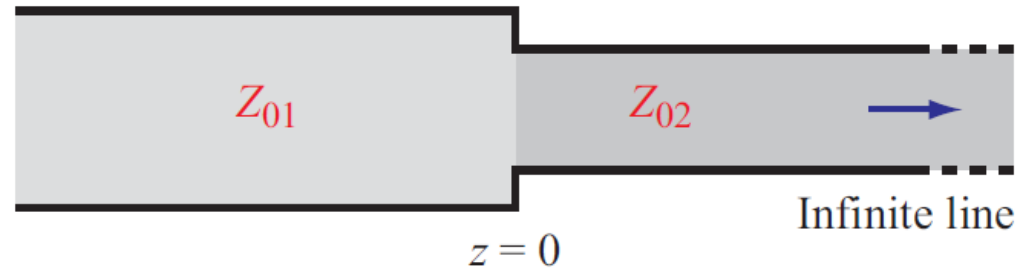
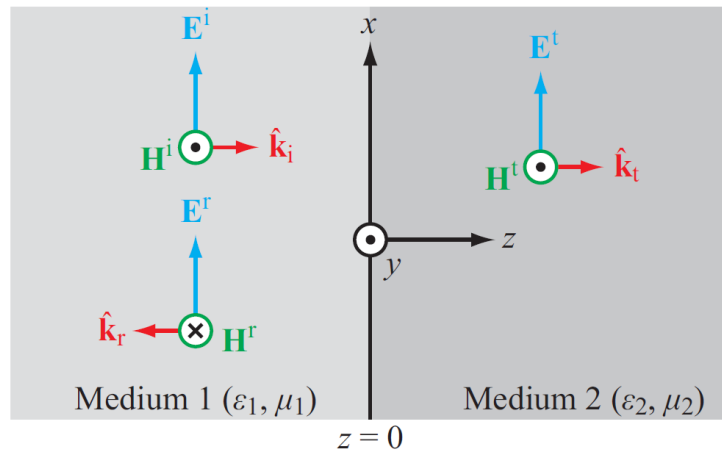


Table 8-1: Analogy between plane-wave equations for normal incidence and transmission-line equations, both under lossless conditions.

Plane Wave [Fig. 8-4(a)]	Transmission Line [Fig. 8-4(b)]
$\tilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^i (e^{-jk_1 z} + \Gamma e^{jk_1 z})$ (8.5a)	$\tilde{V}_1(z) = V_0^+ (e^{-j\beta_1 z} + \Gamma e^{j\beta_1 z})$ (8.5b)
$\tilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \frac{E_0^i}{\eta_1} (e^{-jk_1 z} - \Gamma e^{jk_1 z})$ (8.6a)	$\tilde{I}_1(z) = \frac{V_0^+}{Z_{01}} (e^{-j\beta_1 z} - \Gamma e^{j\beta_1 z})$ (8.6b)
$\tilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E_0^i e^{-jk_2 z}$ (8.7a)	$\tilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2 z}$ (8.7b)
$\tilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \frac{E_0^i}{\eta_2} e^{-jk_2 z}$ (8.8a)	$\tilde{I}_2(z) = \tau \frac{V_0^+}{Z_{02}} e^{-j\beta_2 z}$ (8.8b)
$\Gamma = (\eta_2 - \eta_1) / (\eta_2 + \eta_1)$	$\Gamma = (Z_{02} - Z_{01}) / (Z_{02} + Z_{01})$
$\tau = 1 + \Gamma$	$\tau = 1 + \Gamma$
$k_1 = \omega \sqrt{\mu_1 \epsilon_1}, \quad k_2 = \omega \sqrt{\mu_2 \epsilon_2}$	$\beta_1 = \omega \sqrt{\mu_1 \epsilon_1}, \quad \beta_2 = \omega \sqrt{\mu_2 \epsilon_2}$
$\eta_1 = \sqrt{\mu_1 / \epsilon_1}, \quad \eta_2 = \sqrt{\mu_2 / \epsilon_2}$	Z_{01} and Z_{02} depend on transmission-line parameters

Power Transfer

Medium 1

$$\mathbf{S}_{\text{av}_1}(z) = \frac{1}{2} \Re[\tilde{\mathbf{E}}_1(z) \times \tilde{\mathbf{H}}_1^*(z)]$$

$$= \frac{1}{2} \Re \left[\hat{\mathbf{x}} E_0^i (e^{-jk_1 z} + \Gamma e^{jk_1 z}) \right. \\ \left. \times \hat{\mathbf{y}} \frac{E_0^{i*}}{\eta_1} (e^{jk_1 z} - \Gamma^* e^{-jk_1 z}) \right]$$

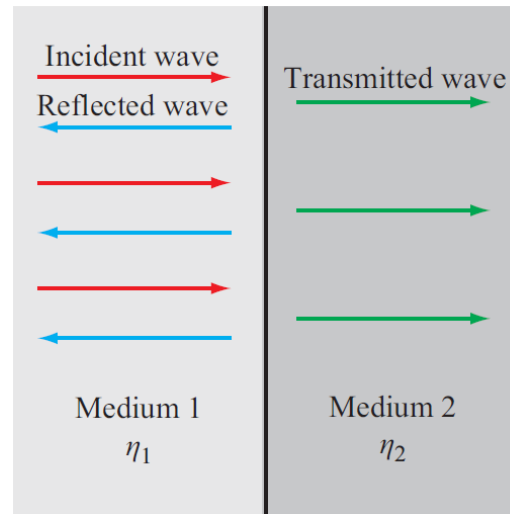
$$= \hat{\mathbf{z}} \frac{|E_0^i|^2}{2\eta_1} (1 - |\Gamma|^2),$$

Medium 2

$$\mathbf{S}_{\text{av}_2}(z) = \frac{1}{2} \Re[\tilde{\mathbf{E}}_2(z) \times \tilde{\mathbf{H}}_2^*(z)]$$

$$= \frac{1}{2} \Re \left[\hat{\mathbf{x}} \tau E_0^i e^{-jk_2 z} \times \hat{\mathbf{y}} \tau^* \frac{E_0^{i*}}{\eta_2} e^{jk_2 z} \right]$$

$$= \hat{\mathbf{z}} |\tau|^2 \frac{|E_0^i|^2}{2\eta_2}.$$



$$\mathbf{S}_{\text{av}_1} = \mathbf{S}_{\text{av}}^i + \mathbf{S}_{\text{av}}^r,$$

$$\mathbf{S}_{\text{av}}^i = \hat{\mathbf{z}} \frac{|E_0^i|^2}{2\eta_1},$$

$$\mathbf{S}_{\text{av}}^r = -\hat{\mathbf{z}} |\Gamma|^2 \frac{|E_0^i|^2}{2\eta_1} = -|\Gamma|^2 \mathbf{S}_{\text{av}}^i.$$

$$\frac{\tau^2}{\eta_2} = \frac{1 - \Gamma^2}{\eta_1} \quad (\text{lossless media}),$$

leads to

$$\mathbf{S}_{\text{av}_1} = \mathbf{S}_{\text{av}_2}.$$

Example 8-2: Yellow Light Incident upon a Glass Surface

A beam of yellow light with wavelength $0.6 \mu\text{m}$ is normally incident in air upon a glass surface. If the surface is situated in the plane $z = 0$ and the relative permittivity of glass is 2.25, determine:

- the standing-wave ratio, and
- the fraction of the incident power transmitted into the glass medium.

Example 8-2 cont.

$$\eta_1 = \sqrt{\frac{\mu_1}{\epsilon_1}} = \sqrt{\frac{\mu_0}{\epsilon_0}} \simeq 120\pi \text{ } (\Omega),$$

$$\eta_2 = \sqrt{\frac{\mu_2}{\epsilon_2}} = \sqrt{\frac{\mu_0}{\epsilon_0}} \cdot \frac{1}{\sqrt{\epsilon_r}} \simeq \frac{120\pi}{\sqrt{2.25}} = 80\pi \text{ } (\Omega),$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{80\pi - 120\pi}{80\pi + 120\pi} = -0.2.$$

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.2}{1 - 0.2} = 1.5.$$

The fraction of the incident power transmitted into the glass medium is equal to the ratio of the transmitted power density, given by Eq. (8.20), to the incident power density, $S_{\text{av}}^i = |E_0^i|^2 / 2\eta_1$:

$$\frac{S_{\text{av}2}}{S_{\text{av}}^i} = \tau^2 \frac{|E_0^i|^2}{2\eta_2} \bigg/ \left[\frac{|E_0^i|^2}{2\eta_1} \right] = \tau^2 \frac{\eta_1}{\eta_2}.$$

In view of Eq. (8.21),

$$\frac{S_{\text{av}2}}{S_{\text{av}}^i} = 1 - |\Gamma|^2 = 1 - (0.2)^2 = 0.96, \text{ or } 96\%.$$

Example 8-3: Normal Incidence on a Metal Surface

A 1-GHz x -polarized plane wave traveling in the $+z$ -direction is incident from air upon a copper surface. The air-to-copper interface is at $z = 0$ and copper has $\epsilon_r = 1$, $\mu_r = 1$, and $\sigma = 5.8 \times 10^7$ S/m. If the amplitude of the electric field of the incident wave is 12 (mV/m), obtain expressions for the instantaneous electric and magnetic fields in the air medium. Assume the metal surface to be several skin depths deep.

Solution: In medium 1 (air), $\alpha = 0$,

$$\beta = k_1 = \frac{\omega}{c} = \frac{2\pi \times 10^9}{3 \times 10^8} = \frac{20\pi}{3} \quad (\text{rad/m}),$$

$$\eta_1 = \eta_0 = 377 \, (\Omega), \quad \lambda = \frac{2\pi}{k_1} = 0.3 \, \text{m}.$$

At $f = 1$ GHz, copper is an excellent conductor because

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0} = \frac{5.8 \times 10^7}{2\pi \times 10^9 \times (10^{-9}/36\pi)} = 1 \times 10^9 \gg 1.$$

Cont.

Example 8-3 (cont.)

$$\begin{aligned}\eta_{c_2} &= (1 + j) \sqrt{\frac{\pi f \mu}{\sigma}} \\ &= (1 + j) \left[\frac{\pi \times 10^9 \times 4\pi \times 10^{-7}}{5.8 \times 10^7} \right]^{1/2} \\ &= 8.25(1 + j) \quad (\text{m}\Omega).\end{aligned}$$

Since η_{c_2} is so small compared to $\eta_0 = 377 \text{ } (\Omega)$ for air, the copper surface acts, in effect, like a short circuit. Hence,

$$\Gamma = \frac{\eta_{c_2} - \eta_0}{\eta_{c_2} + \eta_0} \simeq -1.$$

Cont.

Example 8-3 (cont.)

Upon setting $\Gamma = -1$

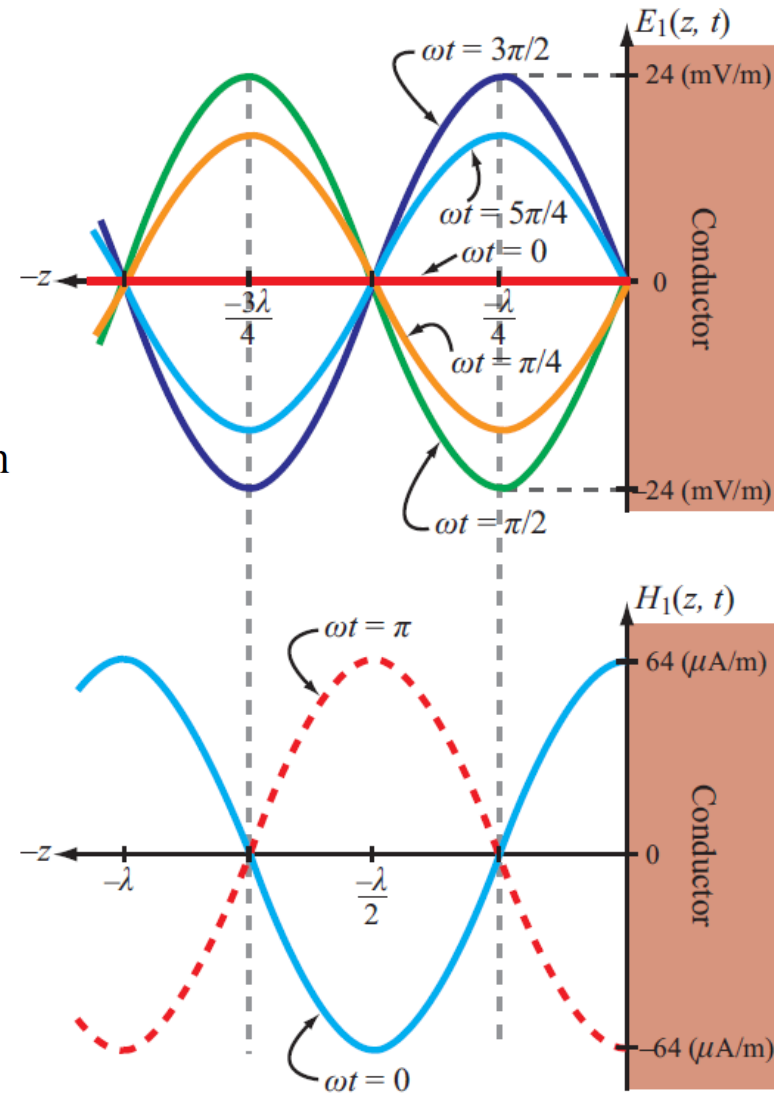
$$\begin{aligned}\tilde{\mathbf{E}}_1(z) &= \hat{\mathbf{x}} E_0^i (e^{-jk_1 z} - e^{jk_1 z}) \\ &= -\hat{\mathbf{x}} j 2 E_0^i \sin k_1 z,\end{aligned}$$

$$\begin{aligned}\tilde{\mathbf{H}}_1(z) &= \hat{\mathbf{y}} \frac{E_0^i}{\eta_1} (e^{-jk_1 z} + e^{jk_1 z}) \\ &= \hat{\mathbf{y}} 2 \frac{E_0^i}{\eta_1} \cos k_1 z.\end{aligned}$$

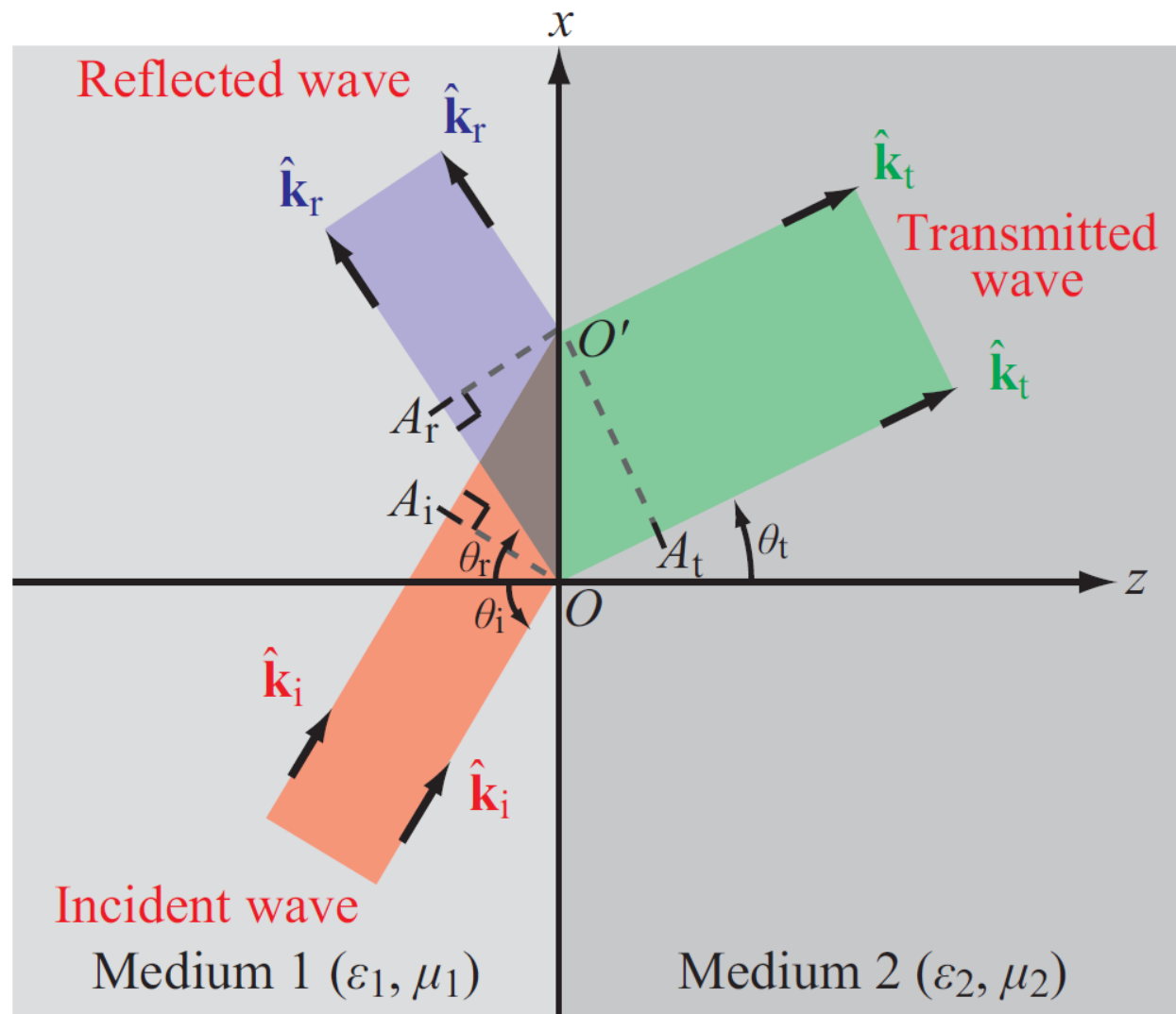
With $E_0^i = 12$ (mV/m), the instantaneous fields associated with these phasors are

$$\begin{aligned}\mathbf{E}_1(z, t) &= \Re[\tilde{\mathbf{E}}_1(z) e^{j\omega t}] \\ &= \hat{\mathbf{x}} 2 E_0^i \sin k_1 z \sin \omega t \\ &= \hat{\mathbf{x}} 24 \sin(20\pi z/3) \sin(2\pi \times 10^9 t) \quad (\text{mV/m}),\end{aligned}$$

$$\begin{aligned}\mathbf{H}_1(z, t) &= \Re[\tilde{\mathbf{H}}_1(z) e^{j\omega t}] \\ &= \hat{\mathbf{y}} 2 \frac{E_0^i}{\eta_1} \cos k_1 z \cos \omega t \\ &= \hat{\mathbf{y}} 64 \cos(20\pi z/3) \cos(2\pi \times 10^9 t) \quad (\mu\text{A/m}).\end{aligned}$$



Snell's Laws



Angles of Incidence, Reflection & Refraction

$$\theta_i = \theta_r \quad (\text{Snell's law of reflection}), \quad (8.28a)$$

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1 \varepsilon_1}{\mu_2 \varepsilon_2}}$$

$$(\text{Snell's law of refraction}). \quad (8.28b)$$

Snell's law of reflection states that the angle of reflection equals the angle of incidence, and Snell's law of refraction provides a relation between $\sin \theta_t$ and $\sin \theta_i$ in terms of the ratio of the phase velocities.

Nonmagnetic Media

Index of refraction n :

$$n = \frac{c}{u_p} = \sqrt{\frac{\mu\epsilon}{\mu_0\epsilon_0}} = \sqrt{\mu_r\epsilon_r} . \quad (8.29)$$

In view of Eq. (8.29), Eq. (8.28b) may be rewritten as

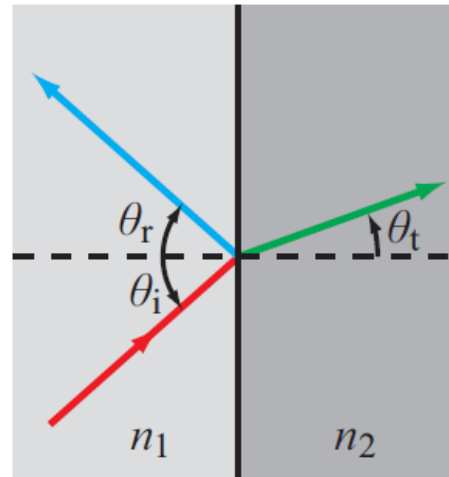
$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\mu_{r1}\epsilon_{r1}}{\mu_{r2}\epsilon_{r2}}} . \quad (8.30)$$

For nonmagnetic materials, $\mu_{r1} = \mu_{r2} = 1$, in which case

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \frac{\eta_2}{\eta_1} \quad (\text{for } \mu_1 = \mu_2). \quad (8.31)$$

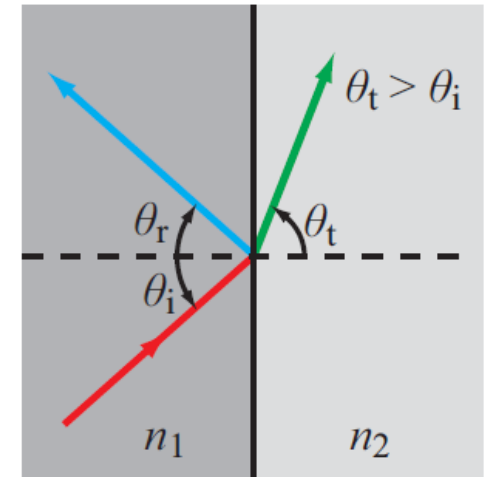
Refraction

Inward refraction



(a) $n_1 < n_2$

Outward refraction

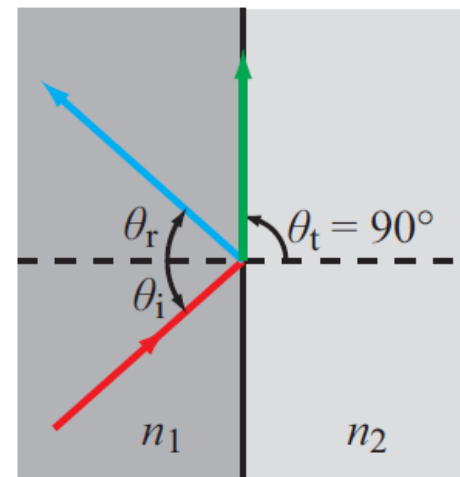


(b) $n_1 > n_2$

When the refraction angle is 90 degrees, the corresponding incidence angle is called the **critical angle**.

$$\begin{aligned}\sin \theta_c &= \frac{n_2}{n_1} \sin \theta_t \Big|_{\theta_t = \pi/2} = \frac{n_2}{n_1} \\ &= \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} \quad (\text{for } \mu_1 = \mu_2).\end{aligned}$$

No transmission



(c) $n_1 > n_2$ and $\theta_i = \theta_c$

Example 8-4: Light Beam Passing through a Slab

A dielectric slab with index of refraction n_2 is surrounded by a medium with index of refraction n_1 , as shown in Fig. 8-11. If $\theta_i < \theta_c$, show that the emerging beam is parallel to the incident beam.

Solution: At the slab's upper surface, Snell's law gives

$$\sin \theta_2 = \frac{n_1}{n_2} \sin \theta_1$$

and, similarly, at the slab's lower surface,

$$\sin \theta_3 = \frac{n_2}{n_3} \sin \theta_2 = \frac{n_2}{n_1} \sin \theta_2.$$

Substituting Eq. (8.33) into Eq. (8.34) gives

$$\sin \theta_3 = \left(\frac{n_2}{n_1} \right) \left(\frac{n_1}{n_2} \right) \sin \theta_1 = \sin \theta_1.$$

Hence, $\theta_3 = \theta_1$. The slab displaces the beam's position, but the beam's direction remains unchanged.

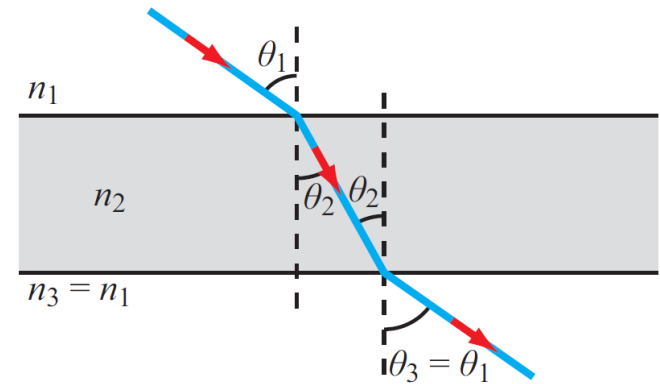


Figure 8-11: The exit angle θ_3 is equal to the incidence angle θ_1 if the dielectric slab has parallel boundaries and is surrounded by media with the same index of refraction on both sides (Example 8-4).

Optical Fiber

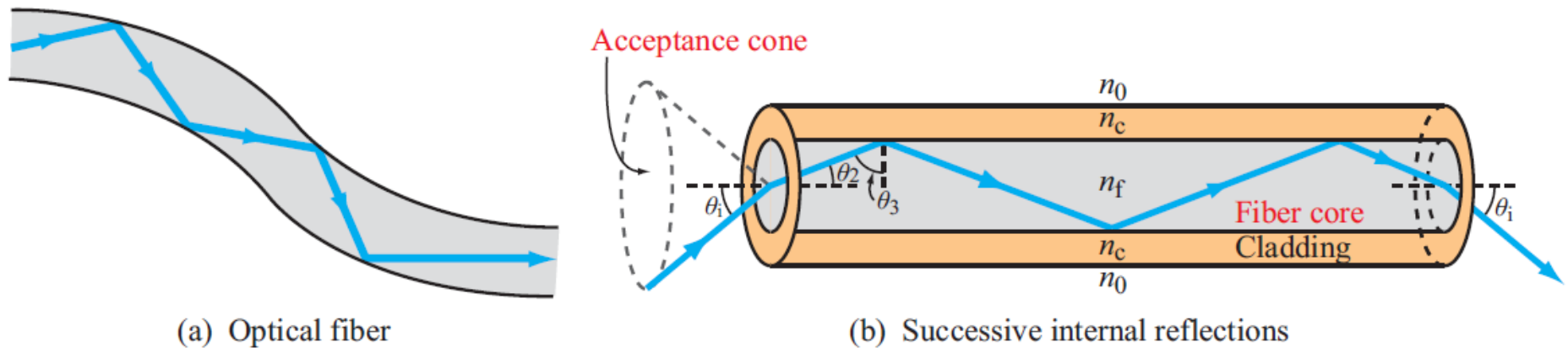


Figure 8-12: Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

The *acceptance angle* θ_a is defined as the maximum value of θ_i for which the condition of total internal reflection remains satisfied:

$$\sin \theta_a = \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2}. \quad (8.40)$$

Modal Dispersion

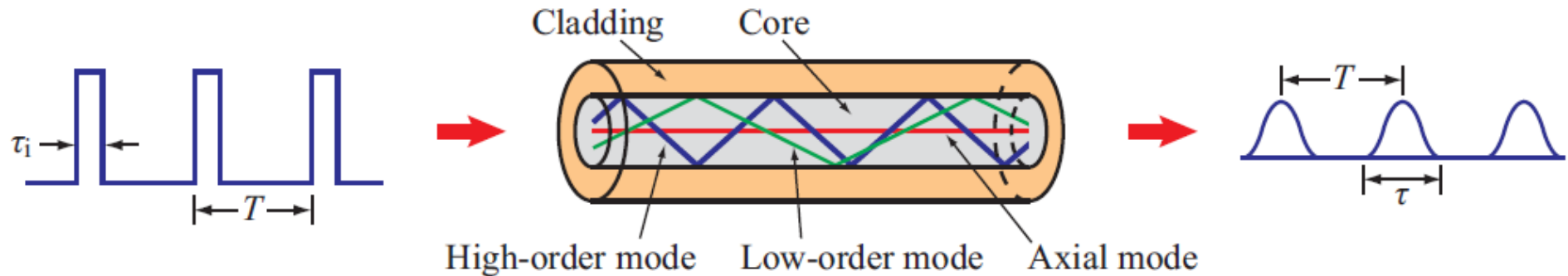


Figure 8-13: Distortion of rectangular pulses caused by modal dispersion in optical fibers.

Any ray of light incident upon the face of the fiber within the acceptance cone can propagate faithfully down the fiber. This means that there can be a large number of ray paths, called *modes*, can travel along the fiber. Higher the angle of rays within the acceptance angle means higher order mode propagation. This means that different modes travel along different paths with different delay time down the fiber. This is called *modal dispersion* with undesirable effect of changing the shape of digital pulses as shown

In order to avoid Intersymbol Interference (ISI) in simple digital NRZ (Non return to zero) transmission, the repetition rate T , and the pulse spread, τ should have the relationship $T \geq 2\tau$.

Highest data rate:

$$f_p = \frac{1}{T} = \frac{1}{2\tau} = \frac{cn_c}{2ln_f(n_f - n_c)} \quad (\text{bits/s})$$

Example 8-5: Transmission Data Rate on Optical Fibers

A 1-km-long optical fiber (in air) is made of a fiber core with an index of refraction of 1.52 and a cladding with an index of refraction of 1.49. Determine

- (a) the acceptance angle θ_a , and
- (b) the maximum usable data rate of signals that can be transmitted through the fiber.

Solution: (a) From Eq. (8.40),

$$\sin \theta_a = \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2} = [(1.52)^2 - (1.49)^2]^{1/2} = 0.3,$$

which corresponds to $\theta_a = 17.5^\circ$.

(b) From Eq. (8.45),

$$\begin{aligned} f_p &= \frac{cn_c}{2ln_f(n_f - n_c)} \\ &= \frac{3 \times 10^8 \times 1.49}{2 \times 10^3 \times 1.52(1.52 - 1.49)} = 4.9 \text{ (Mb/s)}. \end{aligned}$$

8.25 Suppose that the optical fiber of Example 8-5 is submerged in water (with $n = 1.33$) instead of air. Determine θ_a and f_p in that case.

Solution: With $n_0 = 1.33$, $n_f = 1.52$ and $n_c = 1.49$, Eq. (8.40) gives

$$\sin \theta_a = \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2} = \frac{1}{1.33} [(1.52)^2 - (1.49)^2]^{1/2} = 0.23,$$

or

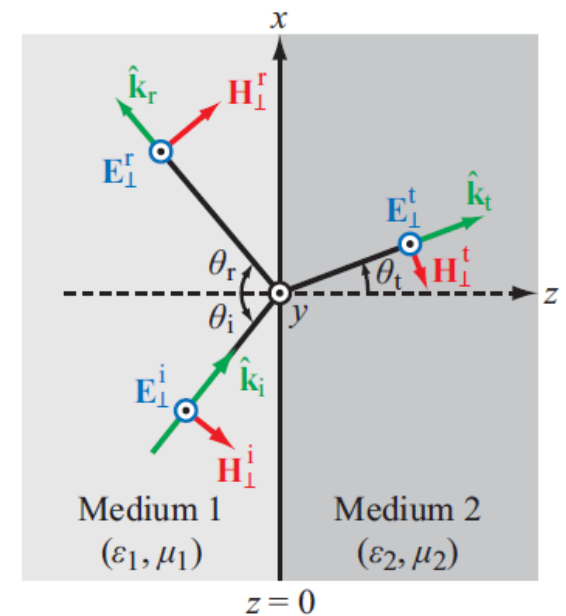
$$\theta_a = 13.1^\circ.$$

The data rate f_p given by Eq. (8.45) is not a function of n_0 , and therefore it remains unchanged at 4.9 (Mb/s).

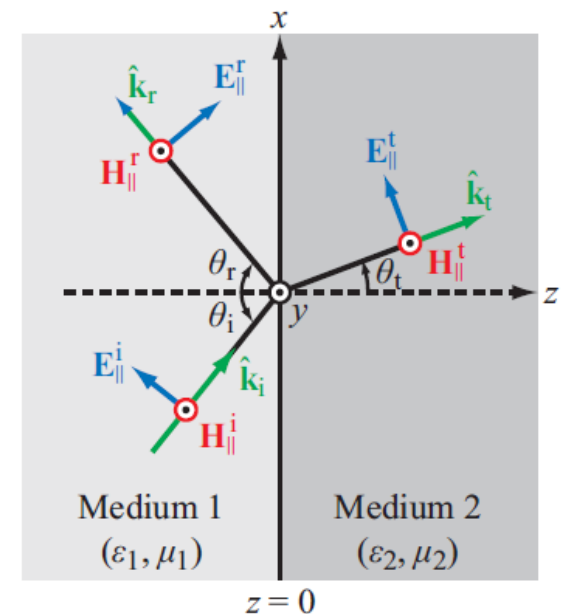
Oblique Incidence

Plane of incidence is defined as the plane containing the normal to the boundary and the direction of propagation of the incident wave (x-y plane in the figure).

A wave of arbitrary polarization may be described as the superposition of two orthogonally polarized waves, one with its electric field parallel to the plane of incidence (**parallel polarization**) and the other with its electric field perpendicular to the plane of incidence (**perpendicular polarization**).



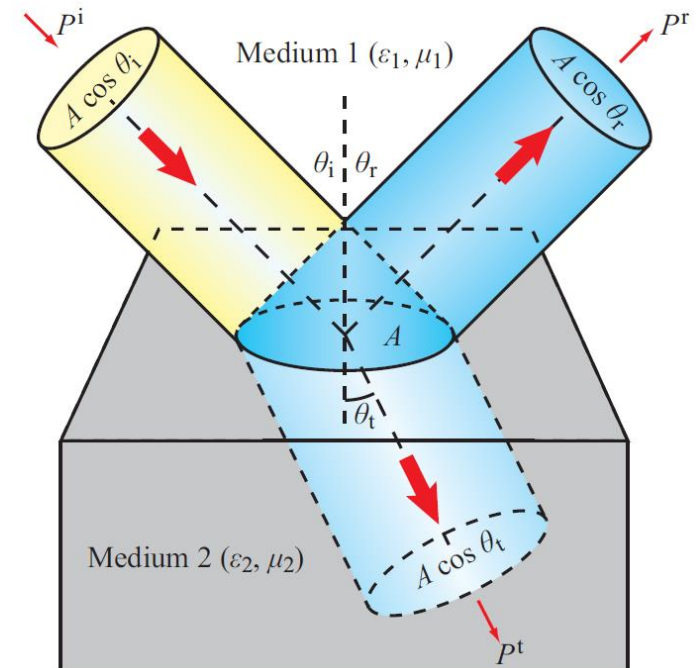
(a) Perpendicular polarization



(b) Parallel polarization

Power Reflectivity and Transmissivity

The incident, reflected, and transmitted waves do not have to obey any such laws as conservation of electric field, conservation of magnetic field, or conservation of power density, but they do have to obey the law of conservation of power.



Summary For Reflection and Transmission

Table 8-2: Expressions for Γ , τ , R , and T for wave incidence from a medium with intrinsic impedance η_1 onto a medium with intrinsic impedance η_2 . Angles θ_i and θ_t are the angles of incidence and transmission, respectively.

Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Transmission coefficient	$\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
Reflectivity	$R = \Gamma ^2$	$R_{\perp} = \Gamma_{\perp} ^2$	$R_{\parallel} = \Gamma_{\parallel} ^2$
Transmissivity	$T = \tau ^2 \left(\frac{\eta_1}{\eta_2} \right)$	$T_{\perp} = \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$T_{\parallel} = \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
Relation of R to T	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$
Notes: (1) $\sin \theta_t = \sqrt{\mu_1 \varepsilon_1 / \mu_2 \varepsilon_2} \sin \theta_i$; (2) $\eta_1 = \sqrt{\mu_1 / \varepsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2 / \varepsilon_2}$; (4) for nonmagnetic media, $\eta_2 / \eta_1 = n_1 / n_2$.			

Summary

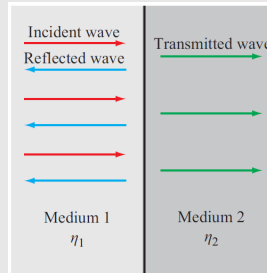
Normal Incidence

$$\Gamma = \frac{E_0^r}{E_0^i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\tau = \frac{E_0^t}{E_0^i} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$\tau = 1 + \Gamma$$

$$\Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} \quad (\text{if } \mu_1 = \mu_2)$$



$$\mathbf{S}_{av1} = \mathbf{S}_{av}^i + \mathbf{S}_{av}^r,$$

$$\mathbf{S}_{av}^i = \hat{\mathbf{z}} \frac{|E_0^i|^2}{2\eta_1},$$

$$\mathbf{S}_{av}^r = -\hat{\mathbf{z}} |\Gamma|^2 \frac{|E_0^i|^2}{2\eta_1} = -|\Gamma|^2 \mathbf{S}_{av}^i.$$

$$\mathbf{S}_{av1} = \mathbf{S}_{av2}.$$

Snell's Laws

$$\theta_i = \theta_r$$

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}}$$

Practice Problems

8.1 A plane wave in air with an electric field amplitude of 20 V/m is incident normally upon the surface of a lossless, nonmagnetic medium with $\epsilon_r = 25$. Determine the following:

- (a) The reflection and transmission coefficients.
- (b) The standing-wave ratio in the air medium.
- (c) The average power densities of the incident, reflected, and transmitted waves.

Solution:**(a)**

$$\eta_1 = \eta_0 = 120\pi \quad (\Omega), \quad \eta_2 = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{120\pi}{5} = 24\pi \quad (\Omega).$$

From Eqs. (8.8a) and (8.9),

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{24\pi - 120\pi}{24\pi + 120\pi} = \frac{-96}{144} = -0.67,$$

$$\tau = 1 + \Gamma = 1 - 0.67 = 0.33.$$

(b)

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.67}{1 - 0.67} = 5.$$

(c) According to Eqs. (8.19) and (8.20),

$$S_{\text{av}}^i = \frac{|E_0^i|^2}{2\eta_0} = \frac{400}{2 \times 120\pi} = 0.52 \text{ W/m}^2,$$

$$S_{\text{av}}^r = |\Gamma|^2 S_{\text{av}}^i = (0.67)^2 \times 0.52 = 0.24 \text{ W/m}^2,$$

$$S_{\text{av}}^t = |\tau|^2 \frac{|E_0^i|^2}{2\eta_2} = |\tau|^2 \frac{\eta_1}{\eta_2} S_{\text{av}}^i = (0.33)^2 \times \frac{120\pi}{24\pi} \times 0.52 = 0.28 \text{ W/m}^2.$$

8.2 A plane wave traveling in medium 1 with $\epsilon_{r1} = 2.25$ is normally incident upon medium 2 with $\epsilon_{r2} = 4$. Both media are made of nonmagnetic, non-conducting materials. If the electric field of the incident wave is given by

$$\mathbf{E}^i = \hat{\mathbf{y}}8 \cos(6\pi \times 10^9 t - 30\pi x) \quad (\text{V/m}).$$

- (a) Obtain time-domain expressions for the electric and magnetic fields in each of the two media.
- (b) Determine the average power densities of the incident, reflected and transmitted waves.

Solution:

(a)

$$\mathbf{E}^i = \hat{\mathbf{y}} 8 \cos(6\pi \times 10^9 t - 30\pi x) \quad (\text{V/m}),$$

$$\eta_1 = \frac{\eta_0}{\sqrt{\epsilon_{r1}}} = \frac{\eta_0}{\sqrt{2.25}} = \frac{\eta_0}{1.5} = \frac{377}{1.5} = 251.33 \, \Omega,$$

$$\eta_2 = \frac{\eta_0}{\sqrt{\epsilon_{r2}}} = \frac{\eta_0}{\sqrt{4}} = \frac{377}{2} = 188.5 \, \Omega,$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{1/2 - 1/1.5}{1/2 + 1/1.5} = -0.143,$$

$$\tau = 1 + \Gamma = 1 - 0.143 = 0.857,$$

$$\mathbf{E}^r = \Gamma \mathbf{E}^i = -1.14 \hat{\mathbf{y}} \cos(6\pi \times 10^9 t + 30\pi x) \quad (\text{V/m}).$$

Note that the coefficient of x is positive, denoting the fact that $\mathbf{E}^r$ belongs to a wave traveling in $-x$ -direction.

$$\mathbf{E}_1 = \mathbf{E}^i + \mathbf{E}^r = \hat{\mathbf{y}} [8 \cos(6\pi \times 10^9 t - 30\pi x) - 1.14 \cos(6\pi \times 10^9 t + 30\pi x)] \quad (\text{A/m}),$$

$$\mathbf{H}^i = \hat{\mathbf{z}} \frac{8}{\eta_1} \cos(6\pi \times 10^9 t - 30\pi x) = \hat{\mathbf{z}} 31.83 \cos(6\pi \times 10^9 t - 30\pi x) \quad (\text{mA/m}),$$

$$\mathbf{H}^r = \hat{\mathbf{z}} \frac{1.14}{\eta_1} \cos(6\pi \times 10^9 t + 30\pi x) = \hat{\mathbf{z}} 4.54 \cos(6\pi \times 10^9 t + 30\pi x) \quad (\text{mA/m}),$$

$$\begin{aligned} \mathbf{H}_1 &= \mathbf{H}^i + \mathbf{H}^r \\ &= \hat{\mathbf{z}} [31.83 \cos(6\pi \times 10^9 t - 30\pi x) + 4.54 \cos(6\pi \times 10^9 t + 30\pi x)] \quad (\text{mA/m}). \end{aligned}$$

Since $k_1 = \omega\sqrt{\mu\epsilon_1}$ and $k_2 = \omega\sqrt{\mu\epsilon_2}$,

$$k_2 = \sqrt{\frac{\epsilon_2}{\epsilon_1}} k_1 = \sqrt{\frac{4}{2.25}} 30\pi = 40\pi \quad (\text{rad/m}),$$

$$\mathbf{E}_2 = \mathbf{E}^t = \hat{\mathbf{y}} 8\tau \cos(6\pi \times 10^9 t - 40\pi x) = \hat{\mathbf{y}} 6.86 \cos(6\pi \times 10^9 t - 40\pi x) \quad (\text{V/m}),$$

$$\mathbf{H}_2 = \mathbf{H}^t = \hat{\mathbf{z}} \frac{8\tau}{\eta_2} \cos(6\pi \times 10^9 t - 40\pi x) = \hat{\mathbf{z}} 36.38 \cos(6\pi \times 10^9 t - 40\pi x) \quad (\text{mA/m}).$$

(b)

$$\mathbf{S}_{\text{av}}^{\text{i}} = \hat{\mathbf{x}} \frac{8^2}{2\eta_1} = \frac{64}{2 \times 251.33} = \hat{\mathbf{x}} 127.3 \quad (\text{mW/m}^2),$$

$$\mathbf{S}_{\text{av}}^{\text{r}} = -|\Gamma|^2 \mathbf{S}_{\text{av}}^{\text{i}} = -\hat{\mathbf{x}} (0.143)^2 \times 0.127 = -\hat{\mathbf{x}} 2.6 \quad (\text{mW/m}^2),$$

$$\begin{aligned} \mathbf{S}_{\text{av}}^{\text{t}} &= \frac{|E_0^{\text{t}}|^2}{2\eta_2} \\ &= \hat{\mathbf{x}} \tau^2 \frac{(8)^2}{2\eta_2} = \hat{\mathbf{x}} \frac{(0.86)^2 64}{2 \times 188.5} = \hat{\mathbf{x}} 124.7 \quad (\text{mW/m}^2). \end{aligned}$$

Within calculation error, $\mathbf{S}_{\text{av}}^{\text{i}} + \mathbf{S}_{\text{av}}^{\text{r}} = \mathbf{S}_{\text{av}}^{\text{t}}$.

8.3 A plane wave traveling in a medium with $\epsilon_{r_1} = 9$ is normally incident upon a second medium with $\epsilon_{r_2} = 4$. Both media are made of nonmagnetic, non-conducting materials. If the magnetic field of the incident plane wave is given by

$$\mathbf{H}^i = \hat{\mathbf{z}} 2 \cos(2\pi \times 10^9 t - ky) \quad (\text{A/m}).$$

- (a) Obtain time-domain expressions for the electric and magnetic fields in each of the two media.
- (b) Determine the average power densities of the incident, reflected, and transmitted waves.

Solution:**(a)** In medium 1,

$$u_p = \frac{c}{\sqrt{\epsilon_{r1}}} = \frac{3 \times 10^8}{\sqrt{9}} = 1 \times 10^8 \quad (\text{m/s}),$$

$$k_1 = \frac{\omega}{u_p} = \frac{2\pi \times 10^9}{1 \times 10^8} = 20\pi \quad (\text{rad/m}),$$

$$\mathbf{H}^i = \hat{\mathbf{z}} 2 \cos(2\pi \times 10^9 t - 20\pi y) \quad (\text{A/m}),$$

$$\eta_1 = \frac{\eta_0}{\sqrt{\epsilon_{r1}}} = \frac{377}{3} = 125.67 \, \Omega,$$

$$\eta_2 = \frac{\eta_0}{\sqrt{\epsilon_{r2}}} = \frac{377}{2} = 188.5 \, \Omega,$$

$$\begin{aligned} \mathbf{E}^i &= -\hat{\mathbf{x}} 2\eta_1 \cos(2\pi \times 10^9 t - 20\pi y) \\ &= -\hat{\mathbf{x}} 251.34 \cos(2\pi \times 10^9 t - 20\pi y) \quad (\text{V/m}), \end{aligned}$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{188.5 - 125.67}{188.5 + 125.67} = 0.2,$$

$$\tau = 1 + \Gamma = 1.2,$$

$$\begin{aligned} \mathbf{E}^r &= -\hat{\mathbf{x}} 251.34 \times 0.2 \cos(2\pi \times 10^9 t + 20\pi y) \\ &= -\hat{\mathbf{x}} 50.27 \cos(2\pi \times 10^9 t + 20\pi y) \quad (\text{V/m}), \end{aligned}$$

$$\begin{aligned} \mathbf{H}^r &= -\hat{\mathbf{z}} \frac{50.27}{\eta_1} \cos(2\pi \times 10^9 t + 20\pi y) \\ &= -\hat{\mathbf{z}} 0.4 \cos(2\pi \times 10^9 t + 20\pi y) \quad (\text{A/m}), \end{aligned}$$

$$\begin{aligned} \mathbf{E}_1 &= \mathbf{E}^i + \mathbf{E}^r \\ &= -\hat{\mathbf{x}} [25.134 \cos(2\pi \times 10^9 t - 20\pi y) + 50.27 \cos(2\pi \times 10^9 t + 20\pi y)] \quad (\text{V/m}), \end{aligned}$$

$$\mathbf{H}_1 = \mathbf{H}^i + \mathbf{H}^r = \hat{\mathbf{z}} [2 \cos(2\pi \times 10^9 t - 20\pi y) - 0.4 \cos(2\pi \times 10^9 t + 20\pi y)] \quad (\text{A/m}).$$

In medium 2,

$$k_2 = \sqrt{\frac{\epsilon_2}{\epsilon_1}} k_1 = \sqrt{\frac{4}{9}} \times 20\pi = \frac{40\pi}{3} \quad (\text{rad/m}),$$

$$\begin{aligned} \mathbf{E}_2 = \mathbf{E}^t &= -\hat{\mathbf{x}} 251.34 \tau \cos\left(2\pi \times 10^9 t - \frac{40\pi y}{3}\right) \\ &= -\hat{\mathbf{x}} 301.61 \cos\left(2\pi \times 10^9 t - \frac{40\pi y}{3}\right) \quad (\text{V/m}), \end{aligned}$$

$$\begin{aligned} \mathbf{H}_2 = \mathbf{H}^t &= \hat{\mathbf{z}} \frac{301.61}{\eta_2} \cos\left(2\pi \times 10^9 t - \frac{40\pi y}{3}\right) \\ &= \hat{\mathbf{z}} 1.6 \cos\left(2\pi \times 10^9 t - \frac{40\pi y}{3}\right) \quad (\text{A/m}). \end{aligned}$$

(b)

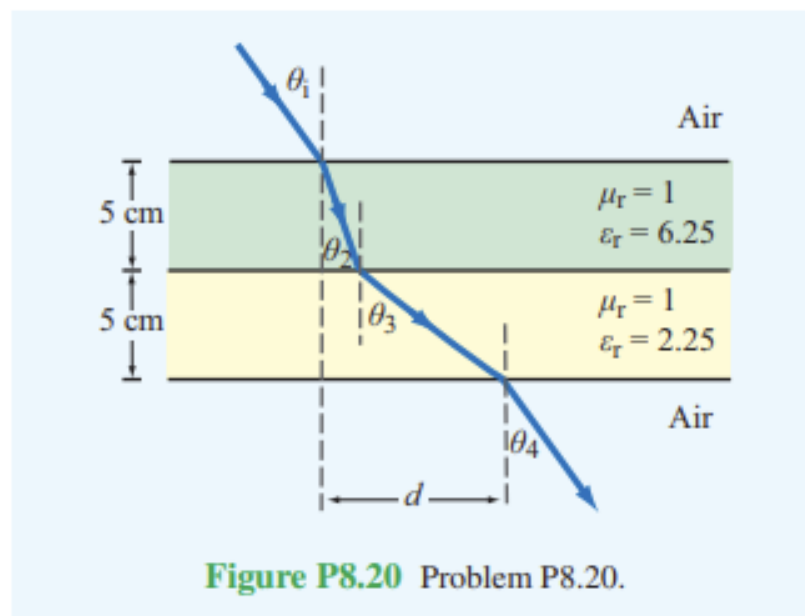
$$\mathbf{S}_{\text{av}}^i = \hat{\mathbf{y}} \frac{|E_0|^2}{2\eta_1} = \hat{\mathbf{y}} \frac{(251.34)^2}{2 \times 125.67} = \hat{\mathbf{y}} 251.34 \quad (\text{W/m}^2),$$

$$\mathbf{S}_{\text{av}}^r = -\hat{\mathbf{y}} |\Gamma|^2 (251.34) = \hat{\mathbf{y}} 10.05 \quad (\text{W/m}^2),$$

$$\mathbf{S}_{\text{av}}^t = \hat{\mathbf{y}} (251.34 - 10.05) = \hat{\mathbf{y}} 241.29 \quad (\text{W/m}^2).$$

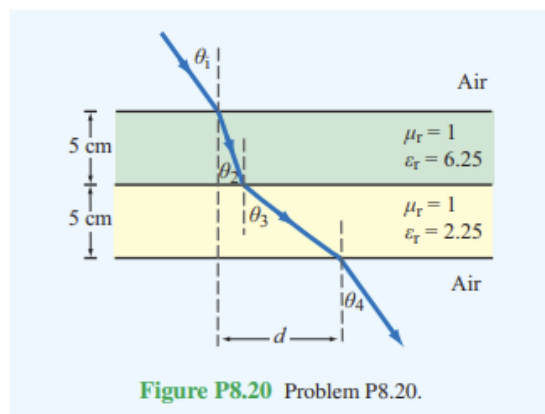
8.20 A parallel-polarized plane wave is incident from air at an angle $\theta_1 = 30^\circ$ onto a pair of dielectric layers as shown in Fig. P8.20.

- (a) Determine the angles of transmission θ_2 , θ_3 , and θ_4 .
- (b) Determine the lateral distance d .



8.20 A parallel-polarized plane wave is incident from air at an angle $\theta_1 = 30^\circ$ onto a pair of dielectric layers as shown in Fig. P8.20.

- (a) Determine the angles of transmission θ_2 , θ_3 , and θ_4 .
 (b) Determine the lateral distance d .



Solution:

- (a) Application of Snell's law of refraction given by (8.31) leads to:

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin 30^\circ \sqrt{\frac{1}{6.25}} = 0.2$$

$$\theta_2 = 11.54^\circ.$$

Similarly,

$$\sin \theta_3 = \sin \theta_2 \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r3}}} = \sin 11.54^\circ \sqrt{\frac{6.25}{2.25}} = 0.33$$

$$\theta_3 = 19.48^\circ.$$

And,

$$\sin \theta_4 = \sin \theta_3 \sqrt{\frac{\epsilon_{r3}}{\epsilon_{r4}}} = \sin 19.48^\circ \sqrt{\frac{2.25}{1}} = 0.5$$

$$\theta_4 = 30^\circ.$$

- (b)

$$d = (5 \text{ cm}) \tan \theta_2 + (5 \text{ cm}) \tan \theta_3$$

$$= 5 \tan 11.54^\circ + 5 \tan 19.48^\circ = 2.79 \text{ cm}.$$

8.21 A light ray incident at 45° passes through two dielectric materials with the indices of refraction and thicknesses given in Fig. P8.21. If the ray strikes the surface of the first dielectric at a height of 2 cm, at what height will it strike the screen?

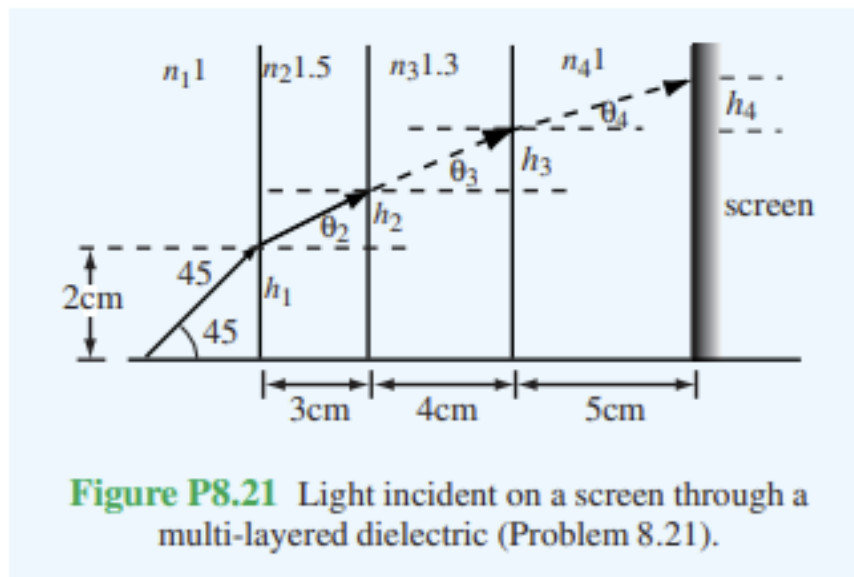
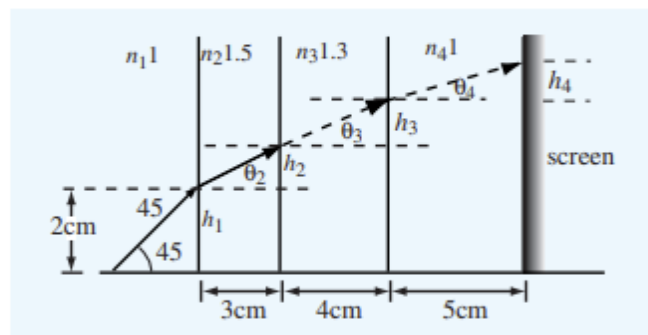


Figure P8.21 Light incident on a screen through a multi-layered dielectric (Problem 8.21).

8.21 A light ray incident at 45° passes through two dielectric materials with the indices of refraction and thicknesses given in Fig. P8.21. If the ray strikes the surface of the first dielectric at a height of 2 cm, at what height will it strike the screen?



Solution:

$$\sin \theta_2 = \frac{n_1}{n_2} \sin \theta_1 = \frac{1}{1.5} \sin 45^\circ = 0.47.$$

Hence,

$$\theta_2 = 28.13^\circ,$$

$$h_2 = 3 \text{ cm} \times \tan \theta_2 = 3 \text{ cm} \times 0.53 = 1.6 \text{ cm},$$

$$\sin \theta_3 = \frac{n_2}{n_3} \sin \theta_2 = \frac{1.5}{1.3} \sin 28.13^\circ = 0.54.$$

Hence,

$$\theta_3 = 32.96^\circ,$$

$$h_3 = 4 \text{ cm} \times \tan 32.96^\circ = 2.6 \text{ cm},$$

$$\sin \theta_4 = \frac{n_3}{n_4} \sin \theta_3 = 0.707.$$

Hence,

$$\theta_4 = 45^\circ,$$

$$h_4 = 5 \text{ cm} \times \tan 45^\circ = 5 \text{ cm}.$$

$$\text{Total height} = h_1 + h_2 + h_3 + h_4 = (2 + 1.6 + 2.6 + 5) = 11.2 \text{ cm}.$$